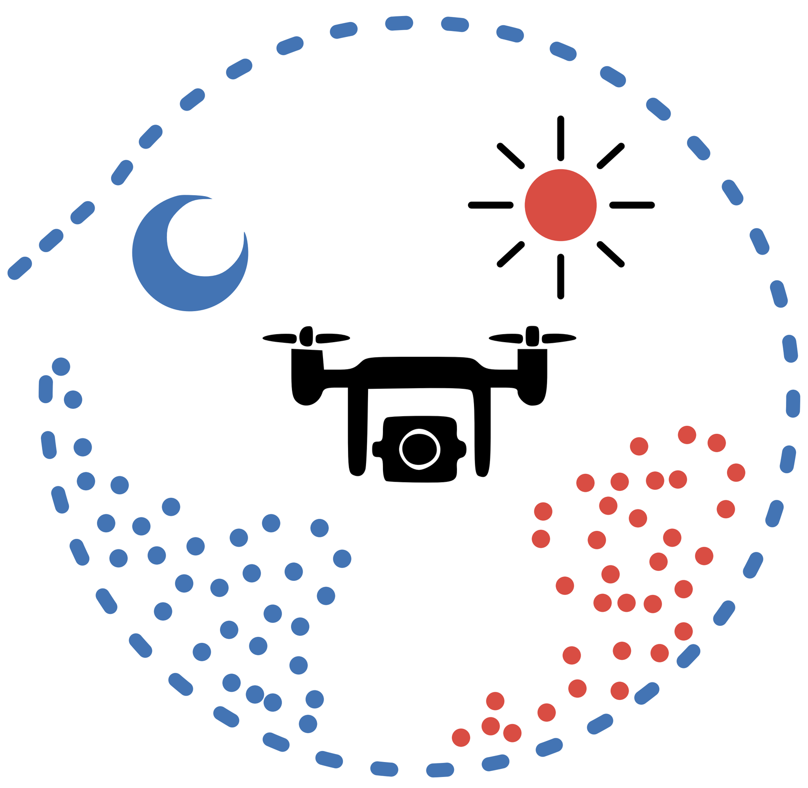
Should UAV-lidar be **collected at night**?

Impacts of Solar Illumination on Intensity, Penetration, and Ground Surface Detection Over Dense and Wet Vegetation



Introduction

UAV lidar is sensitive to environmental conditions that affect signal strength, vegetation penetration, and classification accuracy.

Although nighttime flights reduce solar noise, humidity, dew, and fog may offset these gains. This study examines how illumination influences UAV lidar data quality in peatland vegetation.

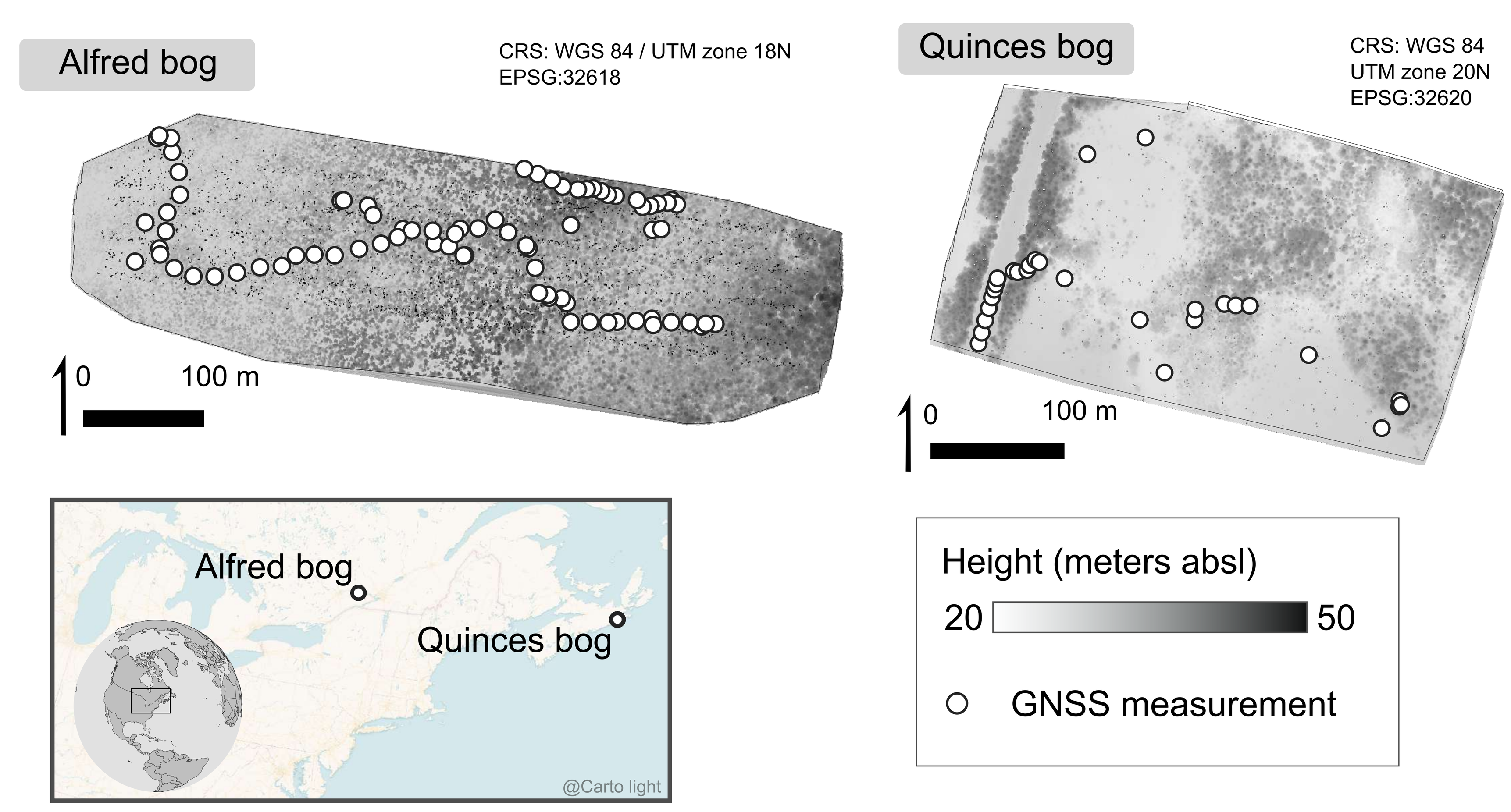
Hypotheses

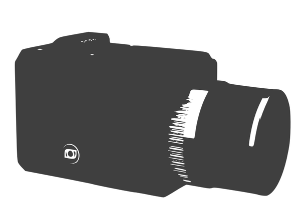
H1: Nighttime UAV lidar **improves data quality** compared with daytime acquisition under variable vegetation and surface wetness conditions.

H2: Nighttime acquisition enables UAV lidar to maintain comparable point-cloud quality at higher flight altitudes than daytime acquisition.

Data

- 12 UAV missions across:
  2 bog sites × 3 flight altitudes × 2 acquisition periods
  Alfred/Quinces × 40/75/100 m × day/night
- 147 ground reference points
- Three surface canopy conditions
- Balkotech Connectiv with Hesai XT32M2x lidar sensor





Methods

1. Define ROI
Setting a common flight mission area in each bog site

2. Retile files
Create an efficient processing grid

3. Compute ROI metrics

4. Classify ground points
Progressive TIN Densification (PTM)

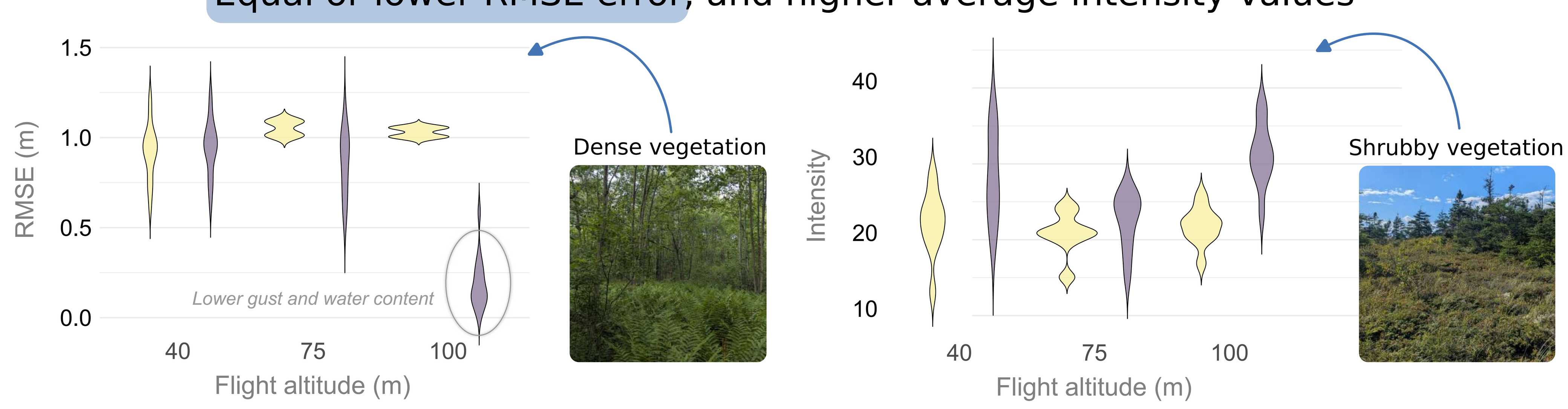
5. Compute target metrics
Use ground points within 50 cm radius and GNSS reference height data

At night...

Fewer negative heights

100 m flights match 75 m daytime 2nd/3th returns

Equal or lower RMSE error, and higher average intensity values



Results and conclusion

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Cristian Iranzo[1], Nicholas Pontone[2], Fernando Pérez-Cabello[1], Jorge Angás[3], Koreen Millard[2]

(1) Universidad de Zaragoza; (2) Carleton University; (3) ARAID - Universidad de Zaragoza